ASSIGNMENT-3

Student Name: Sujie. S

Register Number:732924CSR163

Course Name:BE-CSE

Class:24CS4C

Subject:DATABASE MANAGEMENT SYSTEMS

Assignment Title: SQL Query Practice

Query Execution:

1)create table

```
CREATE TABLE suji_students(
Studentid INT,
name VARCHAR(20),
```

Department VARCHAR(6),
Marks INT,
City VARCHAR(40)
);

2)insert values into tables

INSERT INTO suji_students
VALUES(205,'sri','CSE',76,'POOLAMPATTI');
INSERT INTO suji_students
VALUES(102,'suji','CSE',99,'KONERIPATTI');
INSERT INTO suji_students
VALUES(123,'neha','ECE',78,'AMMAPETTAI');
INSERT INTO suji_students VALUES(110,'selva','EEE',98,'CHENNAI');
INSERT INTO suji_students VALUES(105,'shalini','CSE',90,NULL);
INSERT INTO suji_students VALUES(106,'ruba','ECE',99,'ERODE');
INSERT INTO suji_students VALUES(109,'reyah','CSE',99,'ERODE');

Sample table:

*	Studentid	name	Department	Marks	City
1	205	sri	CSE	76	POOLAMPATTI
2	102	suji	CSE	99	KONERIPATTI
3	123	neha	ECE	78	AMMAPETTAI
4	110	selva	EEE	98	CHENNAI
5	105	shalini	CSE	90	(null)
6	106	ruba	ECE	99	ERODE
7	109	reyah	CSE	99	ERODE

Queries:

1. **Write an SQL query to display all student records from the Student table**

```
SELECT * FROM suji_students;
```

*	Studentid	name	Department	Marks	City
1	205	sri	CSE	76	POOLAMPATTI
2	102	suji	CSE	99	KONERIPATTI
3	123	neha	ECE	78	AMMAPETTAI
4	110	selva	EEE	98	CHENNAI
5	105	shalini	CSE	90	(null)
6	106	ruba	ECE	99	ERODE
7	109	reyah	CSE	99	ERODE

2) Write a query to display the Name and Marks of all students

SELECT name, Marks FROM suji_students;

*	name	Marks
1	sri	76
2	suji	99
3	neha	78
4	selva	98
5	shalini	90
6	ruba	99
7	reyah	99

3) Write a query to display students who scored more than 80 marks

SELECT * FROM suji_students WHERE Marks > 80;

*	Studentid	name	Department	Marks	City
1	102	suji	CSE	99	KONERIPATTI
2	110	selva	EEE	98	CHENNAI
3	105	shalini	CSE	90	(null)
4	106	ruba	ECE	99	ERODE
5	109	reyah	CSE	99	ERODE

4) Write a query to display students belonging to the CSE department

SELECT * FROM suji_students WHERE Department='CSE';

*	Studentid	name	Department	Marks	City
1	205	sri	CSE	76	POOLAMPATTI
2	102	suji	CSE	99	KONERIPATTI
3	105	shalini	CSE	90	(null)
4	109	reyah	CSE	99	ERODE

5) Write a query to display students whose city is Chennai

SELECT * FROM suji_students WHERE City='CHENNAI';

*	Studentid	name	Department	Marks	City
1	110	selva	EEE	98	CHENNAI

6) Write a query to display students whose marks are between 70 and 90

SELECT * FROM suji_students WHERE Marks BETWEEN 70 AND 90;

*	Studentid	name	Department	Marks	City
1	205	sri	CSE	76	POOLAMPATTI
2	123	neha	ECE	78	AMMAPETTAI
3	105	shalini	CSE	90	(null)

7) Write a query to display student names that end with the letter 'a' using the LIKE operator

SELECT name FROM suji_students WHERE name LIKE '%a';

*	name
1	neha
2	selva
3	ruba

8) Write a query to display distinct departments from the Student table

SELECT DISTINCT Department FROM suji_students;

*	Department
1	CSE
2	ECE
3	EEE

9) Write a query to display students whose City value is NULL

SELECT * FROM suji_students WHERE City IS NULL;

*	Studentid	name	Department	Marks	City
1	105	shalini	CSE	90	(null)

10) Write a query to find the highest marks obtained by a student using an aggregate function

SELECT MAX(Marks) AS HighestMarks FROM suji_students;

*	HighestMarks
1	99

11) Write a query to find the average marks of students in the ECE department

SELECT AVG(Marks) AS AverageMarks
FROM suji_students
WHERE Department='ECE';

*	AverageMarks
1	88.5

12) Write a query to count the number of students in each department using GROUP BY

SELECT Department, COUNT(*)
FROM suji_students
GROUP BY Department;

*	Department	COUNT(*)
1	CSE	4
2	ECE	2
3	EEE	1

13) Write a query to display students whose marks are greater than the average marks of all students (nested subquery).

```
SELECT * FROM suji_students
WHERE Marks > (SELECT AVG(Marks) FROM suji_students);
```

*	Studentid	name	Department	Marks	City
1	102	suji	CSE	99	KONERIPATTI
2	110	selva	EEE	98	CHENNAI
3	106	ruba	ECE	99	ERODE
4	109	reyah	CSE	99	ERODE

14) . Write an SQL statement to insert a new student record into the Student table.

```
INSERT INTO suji_students
VALUES(206,'priya','CSE',90,'SALEM');
```

New row will be added to existing table:

7	206	priya	CSE	90	SALEM

15) Write an SQL query to delete the student with StudentID = 205

```
DELETE FROM suji_students
WHERE Studentid=205;
```

*	Studentid	name	Department	Marks	City
1	102	suji	CSE	99	KONERIPATTI
2	123	neha	ECE	78	AMMAPETTAI
3	110	selva	EEE	98	CHENNAI
4	105	shalini	CSE	90	(null)
5	106	ruba	ECE	99	ERODE
6	109	reyah	CSE	99	ERODE
7	206	priya	CSE	90	SALEM

SOURCE:

DB VISUALIZER

SQLite 20000

```sql
CREATE TABLE suji_student(
Studentid INT,
name VARCHAR(20),
Department VARCHAR(6),
Marks INT,
City VARCHAR(40)
);

INSERT INTO suji_student VALUES(205,'sri','CSE',76,'POOLAMPATTI');
INSERT INTO suji_student VALUES(102,'suji','CSE',99,'KONERIPATTI');
INSERT INTO suji_student VALUES(123,'neha','ECE',78,'AMMAPETTAI');
INSERT INTO suji_student VALUES(110,'selva','EEE',98,'CHENNAI');
INSERT INTO suji_student VALUES(105,'shalini','CSE',90,NULL);
INSERT INTO suji_student VALUES(106,'ruba','ECE',99,'ERODE');
```

1 / 1 [638] INS CRLF Auto Commit: ON Cp1252 Untitled*

Log

me	Status	Command	Exec	Fetch	Rows	Message	SQL/Command
09:13:28	INFO					Physical database connection acquired for: SQLite	
09:13:28	SUCCESS	CREATE	0.014		0	OK. No rows were affected	CREATE TABLE suji_student(...
09:13:28	SUCCESS	INSERT	0.007		1	OK	INSERT INTO suji_student VALUES(205,'sri','CSE',76,'POOLAMPATTI')
09:13:28	SUCCESS	INSERT	0.007		1	OK	INSERT INTO suji_student VALUES(102,'suji','CSE',99,'KONERIPATTI')
09:13:28	SUCCESS	INSERT	0.007		1	OK	INSERT INTO suji_student VALUES(123,'neha','ECE',78,'AMMAPETTAI')
09:13:28	SUCCESS	INSERT	0.006		1	OK	INSERT INTO suji_student VALUES(110,'selva','EEE',98,'CHENNAI')
09:13:28	SUCCESS	INSERT	0.007		1	OK	INSERT INTO suji_student VALUES(105,'shalini','CSE',90,NULL)
09:13:28	SUCCESS	INSERT	0.008		1	OK	INSERT INTO suji_student VALUES(106,'ruba','ECE',99,'ERODE')
09:13:29	SUCCESS	INSERT	0.006		1	OK	INSERT INTO suji_student VALUES(109,'reyah','CSE',99,'ERODE')
09:13:29	SUCCESS	SELECT	0.001	0.001	7	Result set fetched	SELECT * FROM suji_student

```sql
SELECT name, Marks FROM suji_student;

SELECT * FROM suji_student WHERE Marks > 80;

SELECT * FROM suji_student WHERE Department='CSE';

SELECT * FROM suji_student WHERE City='CHENNAI';
```

30 / 1 [789] INS CRLF Auto Commit: ON Cp1252 Untitled*

Log

Time	Status	Command	Exec	Fetch	Rows	Message	SQL/Command
09:13:29	SUCCESS	SELECT	0.001	0.001	7	Result set fetched	SELECT name, Marks FROM suji_student
09:13:30	SUCCESS	SELECT	0.001		5	Result set fetched	SELECT * FROM suji_student WHERE Marks > 80
09:13:30	SUCCESS	SELECT	0.001		4	Result set fetched	SELECT * FROM suji_student WHERE Department='CSE'
09:13:31	SUCCESS	SELECT	0.001		1	Result set fetched	SELECT * FROM suji_student WHERE City='CHENNAI'
09:13:31	SUCCESS	SELECT	0.001		3	Result set fetched	SELECT * FROM suji_student WHERE Marks BETWEEN 70 AND 90
09:13:31	SUCCESS	SELECT	0.001		3	Result set fetched	SELECT name FROM suji_student WHERE name LIKE '%a'
09:13:32	SUCCESS	SELECT	0.001		3	Result set fetched	SELECT DISTINCT Department FROM suji_student
09:13:32	SUCCESS	SELECT	0.001	0.001	1	Result set fetched	SELECT * FROM suji_student WHERE City IS NULL
09:13:32	SUCCESS	SELECT	0.001		1	Result set fetched	SELECT MAX(Marks) AS HighestMarks FROM suji_student
09:13:33	SUCCESS	SELECT	0.001	0.003	1	Result set fetched	SELECT AVG(Marks) AS AverageMarks...


```sql
SELECT name, Marks FROM suji_student;

SELECT * FROM suji_student WHERE Marks > 80;

SELECT * FROM suji_student WHERE Department='CSE';

SELECT * FROM suji_student WHERE City='CHENNAI';
```

30 / 1 [789] INS CRLF Auto Commit: ON Cp1252 Untitled*

Log

Time	Status	Command	Exec	Fetch	Rows	Message	SQL/Command
09:13:29	SUCCESS	SELECT	0.001	0.001	7	Result set fetched	SELECT name, Marks FROM suji_student
09:13:30	SUCCESS	SELECT	0.001		5	Result set fetched	SELECT * FROM suji_student WHERE Marks > 80
09:13:30	SUCCESS	SELECT	0.001		4	Result set fetched	SELECT * FROM suji_student WHERE Department='CSE'
09:13:31	SUCCESS	SELECT	0.001		1	Result set fetched	SELECT * FROM suji_student WHERE City='CHENNAI'
09:13:31	SUCCESS	SELECT	0.001		3	Result set fetched	SELECT * FROM suji_student WHERE Marks BETWEEN 70 AND 90
09:13:31	SUCCESS	SELECT	0.001		3	Result set fetched	SELECT name FROM suji_student WHERE name LIKE '%a'
09:13:32	SUCCESS	SELECT	0.001		3	Result set fetched	SELECT DISTINCT Department FROM suji_student
09:13:32	SUCCESS	SELECT	0.001	0.001	1	Result set fetched	SELECT * FROM suji_student WHERE City IS NULL
09:13:32	SUCCESS	SELECT	0.001		1	Result set fetched	SELECT MAX(Marks) AS HighestMarks FROM suji_student
09:13:33	SUCCESS	SELECT	0.001	0.003	1	Result set fetched	SELECT AVG(Marks) AS AverageMarks...


```sql
SELECT * FROM suji_student
WHERE Marks > (SELECT AVG(Marks) FROM suji_student);


INSERT INTO suji_student
VALUES(206,'priya','CSE',90,'SALEM');


DELETE FROM suji_student
WHERE Studentid=205;


SELECT * FROM suji_student;
```

/ 51 [736] INS CRLF Auto Commit: ON Cp1252 Untitled*

Log | 1: suji_student [14] | 2: suji_student [14] | 3: suji_student [11] | 4: suji_student [8] | 5: suji_student [2] | 6: suji_student

ne	Status	Command	Exec	Fetch	Rows	Message	SQL/Command
15:14:36	SUCCESS	SELECT	0.002	0.003	1	Result set fetched	SELECT AVG(Marks) AS AverageMar
15:14:37	SUCCESS	SELECT	0.002	0.002	3	Result set fetched	SELECT Department, COUNT(*) ...
15:14:37	SUCCESS	SELECT	0.001	0.001	8	Result set fetched	SELECT * FROM suji_student...
15:14:38	SUCCESS	INSERT	0.016		1	OK	INSERT INTO suji_student...